Six thin regular geometries for the first Janko group

Human Compression Project

Abstract

We recover the six rank-four geometries of Hartley and Leemans by endpoint conjugation and a central twist. Two subgroup intersections control the four conjugation steps, while explicit coset tests verify flag completion. A complete extension search over proper residues then proves that these are the only such geometries and excludes all higher ranks. The proof retains the full subgroup catalogue of the first Janko group and exact group algorithms, but not a previous catalogue of its geometries. Both required finite programs are given in full in the accompanying appendix.

1 The common statements and finite inputs

The specified source is the ten-page journal article [HL05]. The common paper [U] supplies C.Endpoint, its odd-order consequence C.Odd, the central-twist lemma C.Twist and the exact finite coset criterion C.HellyCount. These provide the group and geometric steps used here. In particular, the intersection property alone is not treated as proof of flag completion for a nonstring diagram.

The mathematical finite inputs are the complete TomLib subgroup catalogue and its permutation model of J_1, exact GAP subgroup and conjugacy algorithms, and a completed Todd–Coxeter calculation. The classical spherical group $[3, 5]$ has order 120. None of these inputs supplies the desired list of geometries.

2 Six geometries and their complete census

This application takes the odd branch of Corollary C.Odd. Retain the GAP TomLib J1 permutation model, of order 175560. The construction certificate constructs a marking a, b, c, d within that model with

$$|ab|, |ac|, |ad|, |bc|, |bd|, |cd| = (5, 2, 2, 3, 2, 5),$$
$$(abc)^5 = ((dcb)^5 a)^3 = 1. \tag{J1}$$

It also completes a Todd–Coxeter calculation for the literal presentation with the four involution relations and (J1): the index over $\langle b, c, d\rangle$ is 1463. That subgroup is a quotient of the finite Coxeter group [3,5] of order 120, so the presented group has order at most $1463 \cdot 120 = 175560$. The found generating tuple in the TomLib model supplies a quotient of that order; hence this is the source presentation of J1. A valid permutation action alone would not prove the upper bound: the completed exact coset enumeration is a charged computational input.

Set $w = (bcd)^5$. The certificate proves that w is a nonidentity central involution of $\langle b, c, d\rangle$. The six source-defined markings and their exact invariants are:

Geometry	Marking	Pair orders ab,ac,ad,bc,bd,cd in that marking	Maximal parabolic orders
Γ_1	a, b, c, d	5,2,2,3,2,5	120,20,20,60
Γ_2	a, wb, c, d	5,2,2,6,2,5	120,20,20,660

Geometry	Marking	Pair orders ab,ac,ad,bc,bd,cd in that marking	Maximal parabolic orders
Γ_3	(a, b, c, c^d)	5,2,2,3,3,5	120,20,60,60
Γ_4	(b^a, b, c, d)	5,3,2,3,2,5	120,120,20,60
Γ_5	(b^a, b, c, c^d)	5,3,3,3,3,5	120,120,60,60
Γ_6	$((wb)^a, wb, c, d)$	5,6,2,6,2,5	120,120,20,660

Order-120 parabolics are $C_2 \times A_5$, order-20 ones are $D_{20} = C_2 \times D_{10}$, order-60 ones A5 and order-660 ones $\mathrm{PSL}_2(11)$; D10 here has order ten. Structure identifications use the retained exact GAP algorithms and named group models.

Corollary C.J1. Each of these six marked coset systems is a thin regular residually connected rank-four geometry with full type-preserving group J1. They are pairwise nonisomorphic, even allowing type permutations. These are [HL05] Theorems 3.2, 4.2, 5.2, 5.4, 6.2 and 7.2, including their generation lemmas.

Proof. Lemma C.Twist gives the central twist $\Gamma_1 \to \Gamma_2$ and preserves its first parabolic. Every endpoint operation has odd endpoint order five, so Corollary C.Odd recovers the same group. The construction certificate verifies the two actual subgroup equalities

$$\langle b, c, d\rangle \cap \langle b, c, d\rangle^a = \langle c, d\rangle, \qquad \langle a, b, c\rangle \cap \langle a, b, c\rangle^d = \langle a, b\rangle. \tag{J2}$$

They have order ten. The second gives Γ_3, the first Γ_4. In Γ_3, $\langle b, c, c^d\rangle = \langle b, c, d\rangle$ because $|cd| = 5$, and $\langle c, c^d\rangle = \langle c, d\rangle$; hence the same first equality gives Γ_5. In Γ_2 its first parabolic is unchanged by the twist, so that equality gives Γ_6. Thus after the seed and twist cases, four IP proofs reduce to the two meets (J2). The construction certificate additionally verifies all 256 ordered marked-subset meets for each of the six markings, so the finite result does not depend on a mistaken reuse of a subgroup identity.

For each marking it checks all sixteen equalities (C3), without presuming that a C-group with a nonstring diagram is a geometry. Lemma C.HellyCount and UC.Geometry now give all the geometric conclusions. The six multisets of maximal parabolic orders in the table are distinct, so no type-permuting isomorphism identifies two geometries. □

The definition of Γ_3 in [HL05] §5 duplicates an A5 parabolic, whereas its printed figure labelled Γ_3 duplicates $C_2 \times A_5$; the definition of Γ_4 has the reverse discrepancy. The source figure labels Γ_3 and Γ_4 are therefore transposed relative to its definitions. The table follows the definitions. This is a source correction, not a seventh construction.

The two source-mentioned failures also follow from the same certificate. The markings (a, wb, c, c^d) and (a,b,c,wd) still generate J1 and satisfy typed Helly, but each has eight failing ordered IP meets; their panel orders are respectively (2,4,2,2) and (2,2,2,4). Therefore they are not thin. These failures demonstrate why neither endpoint generation nor flag transitivity is a substitute for IP.

2.1 Exhaustive extension of proper residues

The original geometry census can be replaced by a short complete search on proper residues. The required group data are the complete TomLib table of J1 subgroup-conjugacy classes, not a list of its geometries.

Theorem C.Census. Let G be finite and let $M_1, \ldots, M_s$ represent all conjugacy classes of maximal proper subgroups. For each M_j enumerate, up to conjugation by M_j, all unordered k-element sets of involutions whose markings satisfy IP and typed Helly. Append every involution of G outside the subgroup generated by each retained set and apply the same tests. This finds every rank-(k+1) thin regular residually connected geometry whose full type-preserving group is a subgroup of G. To restrict to group G, retain exactly the extensions generating G.

Proof. For any such rank-(k+1) geometry, its distinguished marking consists of independent involutions. Omitting one gives a proper subgroup P of its full group and hence of G, so P lies in a conjugate of one M_j. Its marked coset geometry is the corresponding rank-k residue, thus satisfies both IP and typed Helly. Conjugation into M_j and then within M_j puts its unordered markers in the enumerated list. Its omitted involution lies outside P and occurs in the extension loop. Conversely, the tested IP and Helly conditions give exactly the asserted geometries by UC.Geometry. More formally, for each M_j take E to be its admissible unordered k-marker sets and Z to be pairs (T,t) with $T \in E$, t an involution outside $\langle T \rangle$, and the required extension properties. Conjugation by M_j acts on both sets, and $(T, t) \mapsto T$ is equivariant. UC.ExtensionOrbits identifies the extension orbits over a representative T with the stabilizer orbits in its fiber. The certificate tries the whole fiber, so it certainly meets each such orbit; its final conjugacy keys remove duplicate extensions. Ordering the marker set merely orders the type labels. □

For completeness, the residue statement used here requires no extension of arbitrary residue automorphisms to G. The base-flag stabilizer acts transitively and freely on the residue's chambers, and the residue is thin and connected. Its full type-preserving group also acts freely, so the stabilizer already is that full group. Walking inside the residue identifies its panel generators. Thus local Helly is a necessary condition for these genuine residues, not an arbitrary extra filter.

Corollary C.Classification. Up to isomorphism allowing a permutation of types, the six geometries in Corollary C.J1 are all the thin regular residually connected rank-four geometries with group J1. There are none of rank at least five.

Proof. The exhaustive certificate executes Theorem C.Census for $k = 3$ and $k = 4$. The retained complete table has forty subgroup classes. Its nonzero containment entries determine seven maximal classes of orders 660,168,120,114,110,60,42. The local counts are:

Maximal subgroup order	Involutions	Triple orbits	Quadruple orbits	Regular triple orbits	Regular quadruple orbits
660	55	60	588	42	6
168	7	3	3	2	0
120	31	108	616	45	7
114	19	12	42	0	0
110	11	2	4	0	0
60	23	58	221	6	2
42	7	2	2	0	0
Total	—	245	1476	95	15

The finite conjugation action is used here without changing its orbit relation. Conjugation on the actual set of involutions has a permutation image; every element of that image has an ambient conjugating lift, so its induced subset orbits are exactly the needed conjugacy orbits. Faithfulness of this action is not assumed. No local generated subgroup is required to equal its containing maximal subgroup. The 95 retained triples reduce to 70 J1-conjugacy classes. All 100016 possible outside-involution extensions are tested; forty pass with generated group J1, collapsing to precisely six classes. Their maximal-parabolic order multisets are the six distinct ones in Corollary C.J1, which already supplies six pairwise nonisomorphic constructions.

Here is the exact conjugacy test, so equality of numerical invariants is not mistaken for isomorphism. There is one class of 1463 involutions in J1, with centralizer order 120. Fix a_0. For each x in an unordered marker set T choose g_x with $x^{g_x} = a_0$, enumerate $T^{g_x c}$ for every $c \in C_G(a_0)$, and take the least sorted tuple of involution indices. These are exactly the conjugates of T containing a_0: every conjugator carrying x to a_0 differs from g_x on the right by such a c. Consequently keys agree exactly for inner-conjugate unordered markings. There is no assumption that all automorphisms of J1 are inner. The six different parabolic-order multisets prevent any further isomorphism from merging those classes, while the complete six-class upper bound prevents any additional geometry.

For $k = 4$, the fifteen local regular quadruple classes remain fifteen after J1-conjugacy deduplication. All 21352 extensions by an involution outside their generated subgroup fail the full regular-geometry tests. This stage does not discard proper-subgroup targets. If a regular geometry of rank $r \geq 5$ existed for J1, fixing $r-5$ objects of a base chamber would yield a rank-five regular residue with group contained in J1; Theorem C.Census would have found it. Thus all higher ranks are excluded. This does not assert nonexistence of C-group markings that fail typed Helly. □

The source's successful central modifications in §8, when they generate J1 and yield a thin regular geometry, are consequently covered by the same exhaustive classification. The two displayed failures were checked separately above. The construction certificate also verifies the incidental source equality $N_G(\langle a, b, c\rangle) = \langle a, b, c\rangle$. Its initial vertex figure is the universal spherical [3,5] geometry, since that parabolic has the full Coxeter order 120; its facet is the antipodal hemi-dodecahedral quotient of [5,3], since the A5 quotient has order 60 and kernel the unique central order-two subgroup of $C_2 \times A_5$. These conventional spherical identifications use the named classical Coxeter models.

3 Complete finite programs and formal scope

The companion appendix contains two unchanged full programs:

```
phase40_compress_operations_check.g
phase40_compress_operations_exhaustive.g.
```

The first finds the source marking, completes its presentation index, checks all six constructions and both failed operations, and verifies every required subgroup and coset equality. The second derives the seven maximal subgroup classes from all forty catalogue classes, enumerates the local involution sets and every admitted extension, and produces the exact conjugacy keys used above. The geometric census is an output, not a lookup table read by either program.

The retained exact execution used GAP 4.15.1 and TomLib 1.2.11, with the actual Janko data file bound in its record. It checked 1,536 ordered marked-subgroup intersections and 96 coset equalities for the six constructions, all 100,016 rank-four extensions and all 21,352 rank-five extensions. The unchanged runner `phase150_module_C_run.py` permits a separate fresh replay in an empty output directory. Merely checking its saved record is not described as rerunning the census.

The source's six existence theorems, generation lemmas, full rank classification, two failed changes, incidental normalizer and initial spherical residues are all included. The transposition of two figure labels is explicitly separated from its six constructions. No seventh geometry or new historical classification is claimed.

The existing Lean component for endpoint intersections proves a generic mixed-subgroup identity under explicit hypotheses. It does not formalize the complete endpoint criterion, typed coset tests, TomLib catalogue, coset enumeration or this source classification. Those dependencies remain written or computational at the scopes stated here.

References

[U] *Changes of chamber generators and binary translation groups*, the common paper of Topic C.

[HL05] M. I. Hartley and D. Leemans, *On the thin regular geometries of rank four for the Janko group J_1*, Innovations in Incidence Geometry 1 (2005), 181–190.